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Barcelona Is Turning Subway Trains Into Power Stations

Barcelona is using the regenerative braking of its subways to power trains, stations and neighborhood EV chargers. Could New York do it too?

NATALIE DONBACK | GRIST | OCTOBER 7, 2024



(Photo by Ed Lallo / Getty Images via Grist)

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Most of the passengers emerging from the station in Bellvitge, a working-class neighborhood outside Barcelona, have no idea just how innovative the city's subway system is. Using technology not unlike the regenerative braking found in hybrids and electric vehicles, the trains they rode generated some of the power flowing to the EV chargers in the nearby parking lot, the lights illuminating the station, and the escalators taking them to the platforms.

Every time a train rumbles to a stop, the energy generated by all that friction is converted to electricity, which is fed through inverters and distributed throughout the subway system. One-third of that powers the trains; the rest provides juice to station amenities and a growing network of EV chargers.

The ultra-fast charger outside the Bellvitge station is among four *electrolineras* — Spanish for “electric gas stations” — that went up in July. The city's main transit operator, Transports Metropolitans de Barcelona, or TMB, plans to add three more as the project, called MetroCHARGE, expands. “We're trying to take advantage of the power that's already in the metro system and use that spare energy to feed EV chargers on the street,” said Marc Iglesias, head of sustainable mobility at Àrea Metropolitana de Barcelona, a regional agency working with TMB on the project.

Each year, residents and tourists take 440 million trips on Barcelona's subway system, which includes 165 stations linked by 78 miles of track. The transit agency has installed three inverters so far; 13 more are in progress. Once they're all in place by the end of September, it expects regenerative braking to provide 41 percent of the energy needed to power the trains, a renewable source of energy it says will save about [3,885 metric tons](#) of CO2 emissions annually.

Although many cities, including [Vienna](#), [Philadelphia](#), and [São Paulo](#), use regenerative braking to some degree, Barcelona is among the few to use it so extensively and the first to tap it for electric vehicle charging infrastructure. Utilizing energy that is otherwise lost as heat when a train slows can [significantly reduce a transit system's energy consumption](#). (Other efforts, such as optimizing the settings for Barcelona's semi-autonomous trains and using AI to optimize the ventilation in each car, have further reduced energy needs by double-digit margins.)

With the adoption of MetroCHARGE, 33 percent of the energy used by the trains comes from regenerative braking, or enough to power 25 subway stations, said Jordi Picas, who leads the project and is director of metro systems at TMB. In subway systems that don't deploy regenerative braking, “there's so much energy that's not being used, and not only is it lost, it also generates heat that spreads inside the tunnels and increases the temperature,” he said. Since implementing regenerative braking, the temperature in Barcelona's subway system has decreased by 1.8 degrees Fahrenheit.

Last year, transportation electrification [surpassed renewable energy](#) as the world's largest category of energy transition investment, receiving about \$634 billion globally. Although implementing MetroCHARGE has cost about \$8.6 million (7.8 million euros), TMB expects to recoup that in [four to five years](#) through energy savings and revenue from the charging stations, where drivers pay about 33 cents per kilowatt-hour that flows into their cars.

Metro systems worldwide already have the electric infrastructure needed to adopt this approach, but not all of them use trains outfitted with regenerative braking — and retrofitting it is expensive, Picas said. All of Barcelona's trains have featured the technology since the 1980s. Given that a single train costs about \$6.6 million and has an average lifespan of 35 to 45 years, it's essential that transit operators include them in medium- and long-term planning, he said.

There are other challenges beyond cost, like finding optimal spots for the inverters and charging stations in a dense metropolitan area. “The hardest challenge was reaching an agreement with the various city halls to get access to the public space to set up the chargers,” said Picas.

Other cities have expressed an interest in replicating MetroCHARGE, and TMB recently met a delegation from New Delhi. It has also shared information with officials from Vienna and an international consortium

of 45 transit systems called The Community of Metros Benchmarking Group. Cities like New York — which has the world’s fifth-largest metro system, with 472 stations and 665 miles of track — could see significant energy savings from regenerative braking due to the sheer scale of its subway network, said Ahmed Mohamed, director of graduate studies at the Department of Electrical Engineering at the City College of New York.

[In a 2018 study](#), Mohamed and his team found that the Metropolitan Transit Authority, or MTA, which runs New York’s subways, could cut its energy consumption by 35 percent if it adopted regenerative braking system-wide and used the electricity it generates to power trains and station amenities. As of 2022, just half of the city’s trains use the technology, according to the New York State Energy Research and Development Authority, although any new trains are required to have it. However, “there’s not necessarily a strategic plan for how they can be used for energy saving,” Mohamed said, adding that the MTA, which would not make anyone available to comment, favors regenerative braking because it requires less maintenance than conventional friction brakes.

One of the main barriers to implementing regenerative braking is the lack of data outlining just how much energy, and money, might be saved with its adoption, Mohamed said. “When you’re not quite sure about the savings, it’s hard to run a cost-benefit analysis, so decision-making is not very easy,” he said. “It’s important to fund pilot projects to get the real numbers.”

Learning from projects like MetroCHARGE could help other cities understand the benefits of regenerative braking, Mohamed said. Improved modeling and encouraging transit systems to share the details of their systems so others may learn from them would help, too.

Another challenge is the number of stakeholders involved — including technology and electricity companies — makes determining things like who should pay for the project difficult. “Who’s running it? Who’s controlling it? And what’s the role of different stakeholders?” Mohamed said. He’d like to see the MTA take the lead in adopting this energy-saving technology so the trains rumbling into its stations might one day power cars on the busy New York city streets above.

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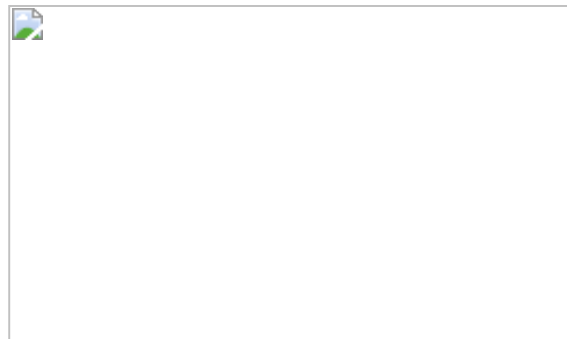
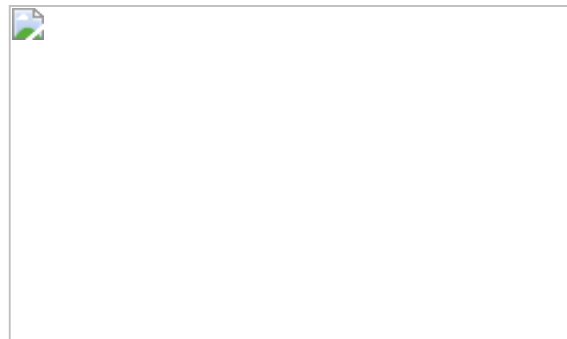


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